Metoclopramide increases gastric motility through three main actions that work together

1. D2 receptor antagonism (the main mechanism)

- Dopamine, acting on D2 receptors on enteric neurons, normally inhibits gastric motility. It does this by suppressing acetylcholine (ACh) release from myenteric (cholinergic) neurons.
- Metoclopramide blocks these D2 receptors, which removes the inhibition.
- The result is more ACh release, stronger smooth muscle contraction, and increased gastric tone and peristalsis.

2. 5-HT4 receptor agonism (prokinetic action)

- Metoclopramide stimulates 5-HT4 receptors on enteric cholinergic neurons.
- This activates adenylyl cyclase, raises cAMP, and enhances further ACh release.
- This adds to the prokinetic effect, especially in the upper GI tract.

3. Weak 5-HT3 antagonism

- This contributes mainly to the antiemetic effect (acting at the chemoreceptor trigger zone and vagal afferents). It is not the main driver of motility.

Net effects on the GI tract

- Increased lower oesophageal sphincter (LES) tone, which reduces reflux
- Increased gastric contractions and antral motility
- Relaxation of the pylorus and duodenum, which coordinates antral-duodenal activity
- Faster gastric emptying and shorter intestinal transit
- No major effect on colonic motility

Chelation therapy

Definition

Chelation therapy uses chelating agents to treat heavy metal poisoning or metal overload. A chelator is a ligand with two or more donor groups (S, O, N) that binds a metal ion to form a stable, water-soluble, ring-shaped complex (a chelate). The complex is then excreted, mainly in urine.

Mechanism

- The chelator binds the free metal in plasma and tissues, which removes it from its toxic binding sites (enzymes, proteins).
- The chelate is less toxic and more water-soluble than the free metal.
- The metal-chelator complex is renally excreted (or via bile or faeces for some agents).

Properties of an ideal chelator

- Higher affinity for the toxic metal than for essential metals (e.g. Ca^{2+}, Zn^{2+})
- Forms a non-toxic, water-soluble complex
- Reaches the same compartments as the metal (intracellular or extracellular)
- Oral activity, low toxicity, and no redistribution of metal to sensitive organs such as the brain

Main chelating agents

Agent	*Used for*	*Route*	*Key adverse effects / notes*
Dimercaprol (BAL)	Arsenic, mercury, gold, lead (with EDTA)	Deep IM (in peanut oil)	Hypertension, tachycardia, pain at injection site, haemolysis in G6PD deficiency. Not for iron or cadmium (the complex is toxic)
Calcium disodium EDTA	Lead	IV/IM	Nephrotoxicity. Must be the calcium salt: Na_2EDTA causes severe hypocalcaemia
Succimer (DMSA)	Lead (esp. children), arsenic, mercury	Oral	GI upset, transient raised liver enzymes
Penicillamine	Wilson's disease (copper); also lead, gold	Oral	Nephrotic syndrome, SLE-like syndrome, skin reactions, bone marrow suppression
Trientine	Wilson's disease (if penicillamine not tolerated)	Oral	Fewer side effects than penicillamine
Deferoxamine	Iron overload or acute iron poisoning	IV/SC/IM	Pink-red (vin rosé) urine, hypotension if given fast, ototoxicity, retinal toxicity, *Yersinia* infection risk

Deferasirox / Deferiprone	Chronic iron overload (e.g. thalassaemia)	Oral	Deferasirox: renal and hepatic toxicity. Deferiprone: agranulocytosis
Prussian blue	Thallium, radioactive caesium	Oral	Traps metal in the gut and prevents reabsorption

Eg. Quinolones with milk

- Milk and other dairy products reduce the oral absorption of fluoroquinolones (ciprofloxacin, norfloxacin, ofloxacin, levofloxacin).
- Quinolones have a 4-keto group and a 3-carboxyl group that act as a chelating site.
- In the gut, they bind polyvalent cations (Ca^{2+} from milk, also Mg^{2+}, Al^{3+}, Fe^{2+}/Fe^{3+}, Zn^{2+}) to form insoluble, non-absorbable chelate complexes.
- Less free drug is absorbed, so bioavailability and peak plasma concentration (Cmax) fall.
- Subtherapeutic plasma levels, which can lead to treatment failure
- Increased risk of bacterial resistance

Other substances that interact the same way

- Antacids (Al^{3+}, Mg^{2+}, Ca^{2+}), which have the strongest effect
- Sucralfate
- Iron, zinc, and calcium supplements
- Multivitamins with minerals
- Calcium-fortified juices and yoghurt

Fatty food slows gastric emptying

- Fat (especially long-chain fatty acids) reaching the duodenum and jejunum is detected by enteroendocrine cells.
- These cells release cholecystokinin (CCK), along with GIP, secretin, and peptide YY (the “ileal brake” when fat reaches the distal gut).
- CCK acts on CCK-A receptors on vagal afferent fibres, triggering a vagovagal reflex (the enterogastric reflex) that relaxes the proximal stomach and reduces antral contractions.

- CCK also contracts the pyloric sphincter, which restricts the passage of chyme into the duodenum.
- The net result is slower gastric emptying, which gives time for fat digestion and absorption (bile and pancreatic lipase need time to emulsify and break down fat).

Why fat has the strongest effect

- Fat is the most potent inhibitor of gastric emptying among the macronutrients, followed by protein and then carbohydrate.
- Emptying also slows with acidic, hypertonic, and high-calorie chyme (sensed by duodenal receptors).

P-glycoprotein (P-gp)

What it is

- P-glycoprotein (also called MDR1 or ABCB1) is an ATP-dependent efflux transporter belonging to the ABC (ATP-binding cassette) superfamily.
- It is encoded by the *ABCB1* (MDR1) gene.
- It acts as a membrane "pump" that moves a wide range of drugs and xenobiotics out of cells, against their concentration gradient, using ATP hydrolysis.

Mechanism

- The drug enters the cell membrane or cytoplasm, and P-gp binds it from within the lipid bilayer.
- ATP binding and hydrolysis cause a conformational change, and the drug is expelled into the extracellular space or lumen.
- The result is lower intracellular drug concentration, so less absorption and less tissue penetration.

Location and function

Site	***Function***	***Effect on drugs***
Intestinal epithelium (apical)	Pumps drug back into the gut lumen	Reduces oral bioavailability

Liver (canalicular membrane)	Secretes drug into bile	Increases biliary elimination
Kidney (proximal tubule, apical)	Secretes drug into urine	Increases renal elimination
Blood-brain barrier (capillary endothelium)	Pumps drug back into blood	Limits CNS entry
Placenta	Pumps drug back to maternal circulation	Protects the foetus
Testes	Blood-testis barrier	Protects germ cells

Common substrates

- Digoxin, cyclosporine, tacrolimus
- Anticancer drugs (doxorubicin, vincristine, vinblastine, paclitaxel, etoposide)
- Protease inhibitors (ritonavir, saquinavir)
- Loperamide, fexofenadine, dabigatran, colchicine

Inhibitors (raise substrate levels)

- Verapamil, quinidine, amiodarone, clarithromycin, itraconazole, ritonavir, cyclosporine
- Grapefruit juice (some effect)

Inducers (lower substrate levels)

- Rifampicin, St John's wort, carbamazepine, phenytoin

Clinical relevance

- **Drug interactions:** quinidine or verapamil with digoxin inhibits P-gp, so digoxin levels rise and toxicity can occur. Rifampicin induces P-gp, so digoxin levels fall.
- **Multidrug resistance (MDR) in cancer:** tumour cells overexpress P-gp, pump out chemotherapy drugs, and become resistant to several unrelated agents at once.
- **CNS effects:** loperamide normally cannot enter the brain because of P-gp. With a P-gp inhibitor (e.g. quinidine), central opioid effects can appear.
- **Pharmacokinetics:** it works together with CYP3A4. They share many substrates and inhibitors, so both reduce oral bioavailability.

- **Genetic variation:** *ABCB1* polymorphisms (e.g. C3435T) can change expression and drug levels.

Summary flow

Drug enters the cell → P-gp binds it → ATP hydrolysis → drug effluxed out → reduced absorption, reduced tissue entry, and increased elimination.

Exam point: P-gp is an efflux pump (not an uptake transporter). Its inhibition raises drug levels, and its induction lowers them.

Grapefruit juice and P-gp

- Grapefruit juice contains furanocoumarins (bergamottin, 6',7'-dihydroxybergamottin) and flavonoids (naringin, naringenin) that have been shown to inhibit P-gp in vitro.
- Furanocoumarins irreversibly (mechanism-based) inhibit intestinal CYP3A4.
- Normally, a CYP3A4 substrate (e.g. simvastatin) is partly broken down in the gut wall during absorption, then again in the liver (first-pass metabolism). Little of the dose reaches the circulation.
- With grapefruit juice, the gut-wall step is lost. More drug gets through unchanged, so bioavailability and plasma levels rise (Cmax and AUC go up).
- Higher levels can mean toxicity (e.g. rhabdomyolysis with simvastatin).

CKD's effect on albumin and drug binding

- In cdk and cirrhosis, albumin is reduced, less protein binding, increases free drug, can lead to toxicities.
- **Cirrhosis:** the liver makes albumin, so synthesis falls. Hyperbilirubinaemia also competes with drugs for binding sites.
- **CKD:** albumin is lost in urine (proteinuria, nephrotic syndrome), and malnutrition and chronic inflammation lower it further.
- **CKD has an extra effect:** uraemic toxins and accumulated metabolites compete for binding sites and can change albumin's structure. Binding of acidic drugs falls even when albumin is not very low (e.g. phenytoin).
- Less binding means a higher **free fraction**, so a given total concentration has a stronger effect and carries a higher risk of toxicity.
- Drugs that are **highly protein-bound (>90%)**, have a **narrow therapeutic index**, and have a **small Vd**

- Examples: phenytoin, warfarin, valproate, and some NSAIDs